

PUBLISHED BY THE AGENTICS FOUNDATION

The OIA Reference Model

A structural reference for the design, deployment, and governance of enterprise intelligent systems.

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Version 0.5 · 2026

WHY THIS MATTERS NOW

Reference architectures matter most when the technology is moving faster than the organisations deploying it.

We are in such a moment now.

Enterprise intelligent systems — agents, models, and automation that increasingly do real work alongside human teams — are arriving inside our companies before the architectural vocabulary to govern them has stabilised.

Boards are asking strategic questions their management teams cannot yet answer in structural terms. CIOs are committing capital before the architecture is drawn. CISOs are defending against threat surfaces existing frameworks do not name.

PART ONE

The OIA Reference Model

The structural shape of enterprise intelligent systems.

Three properties define enterprise intelligent systems.

PERSISTENCE

Intelligent systems maintain state across interactions.

A conversation continues across sessions. An agent's knowledge of the organisation accumulates. A model's behaviour reflects its training history. Persistence is not optional — it is what makes the system useful, and what makes it harder to govern.

Layers 8 & 3

AUTONOMY

Intelligent systems act with degrees of autonomy past applications could not.

They take multi-step actions. They invoke tools. They make decisions inside scopes of authority that humans did not pre-approve case-by-case. They can act faster than humans can supervise them in the moment.

Layer 7

CONSEQUENCE

Outputs and actions whose correctness must be demonstrable, against current threat reality.

A patched library, a swapped model, a new jailbreak — each changes correctness mid-flight. Architectural posture is time-bound. Consequence is governed continuously, not at release.

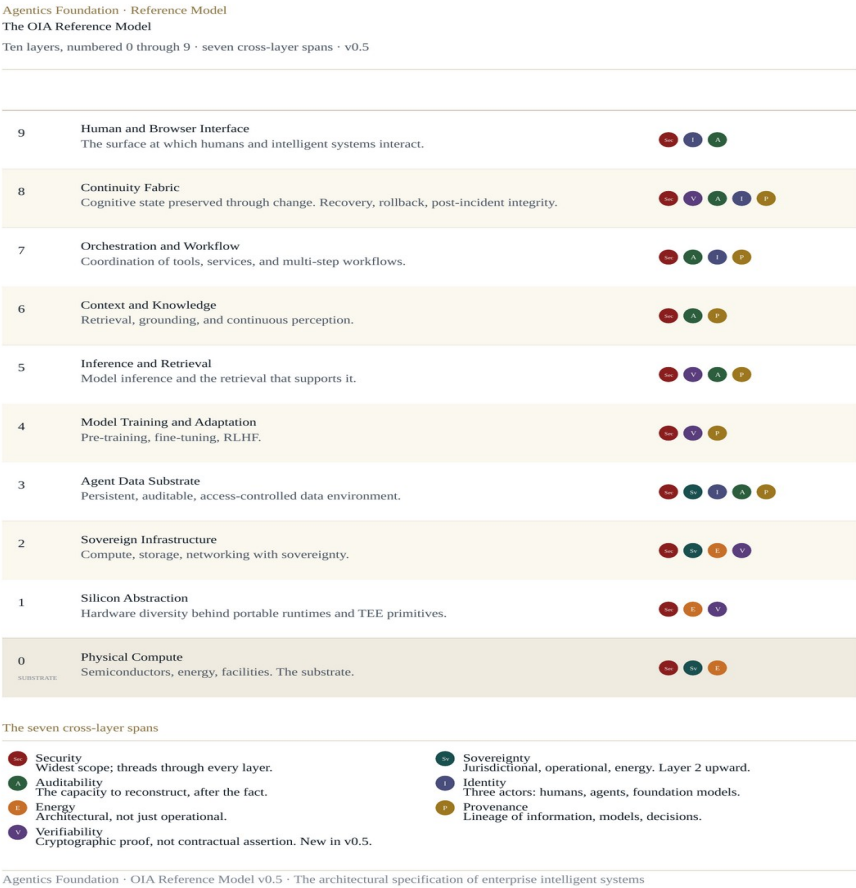
All seven cross-layer spans

Ten layers, numbered 0 through 9. Seven cross-layer spans.

Layer 0 is the physical substrate.
Layers 1–9 are the architectural stack.

Two layers hold state: Layer 3 (substrate) and Layer 8 (fabric). The pairing is asymmetric — substrate below, fabric above the operational stack — and architecturally deliberate.

Seven spans thread vertically: Security, Sovereignty, Auditability, Identity, Energy, Provenance, and — new in v0.5 — Verifiability.



The Continuity Fabric is the layer the rest of the architecture is paired against.

Cognitive state preserved through change. Recovery, rollback, post-incident integrity.

No agent framework, model provider, or orchestration platform can be allowed to own the enterprise's cognitive state — because all of them will eventually change.

Layer 8 is the layer most consistently identified across audiences as the gap the existing technology stack does not yet fill:

- CISOs feel it as recovery and rollback gap.
- CIOs feel it as state-coherence gap across agent frameworks.
- Boards feel it as audit-chain integrity gap.

One layer. Three different audience readings. The same architectural absence.

Seven cross-layer spans thread through the model.

Security	Widest scope. Threads through every layer.	Sovereignty	Three dimensions: jurisdictional, operational, energy.
Auditability	The capacity to reconstruct, after the fact.	Identity	Three actors: humans, agents, foundation models.
Energy	Architectural, not just operational.	Provenance	Lineage of information, models, decisions, actions.
Verifiability	Cryptographic proof, not contractual assertion. New in v0.5.		

THE AGENTIC ERA THREAT SURFACES

Four named threats land inside the operational stack — not at the perimeter.

Mimicry

Telling humans from agents — and AI from real people — at every entry point.

Agent-in-the-middle

Adversarial intermediary between agent and tool. The agentic SSL strip.

Agent spoofing

Agent identities inheriting trust they should not have.

Model compromise

The trusted model that is no longer trustworthy. Weight swap, alignment drift, jailbreak.



PART TWO

The OIA Lens System

How the architecture reaches the executives accountable for AI.

THE ARCHITECTURE ALONE IS NOT ENOUGH

Reference architectures become foundational only when audiences can find themselves in them.

The OIA is intentionally neutral. Vendor-neutral. Audience-neutral. That neutrality is what gives the architecture its durability.

It also creates a problem.

Most executives accountable for AI do not work from structural specifications. They work from their own frameworks, their regulatory environment, the questions on their desk this quarter.

Without translation, the OIA risks being filed as a security document by the first CISO who picks it up — or as a strategy paper that never reaches the engineers who must build to it.

The Lens System is the translation.

The architecture is invariant. The overlay changes per audience.

Agentics Foundation · Reference Model		
The Lens Overlay		
How audience-specific lenses read against the architecture. The architecture is invariant; the overlay changes per audience.		
Audience question (illustrative · rotated)	OIA layer · invariant	Lens axes
CISO "Can we tell humans from agents at every entry point?" 9	Human and Browser Interface The surface at which humans and intellige...	Threat axis Operational axis
Board "How do we recover when an autonomous agent has mis-acted?" 8	Continuity Fabric Cognitive state preserved through change....	Foundational axis
CIO "How do we govern eight agent frameworks across our BUs?" 7	Orchestration and Workflow Coordination of tools, services, and mult...	Operational axis
CISO "Can adversaries poison our retrieval or context?" 6	Context and Knowledge Retrieval, grounding, and continuous perc...	Threat axis
Board "How do we audit a model decision after the fact?" 5	Inference and Retrieval Model inference and the retrieval that su...	Foundational axis
CIO "Build, fine-tune, or rely on third-party inference?" 4	Model Training and Adaptation Pre-training, fine-tuning, RLHF.	Operational axis
CISO "What sensitive data is reaching LLM APIs via SaaS?" 3	Agent Data Substrate Persistent, auditable, access-controlled ...	Threat axis
CIO "Where can regulated AI workloads physically run?" 2	Sovereign Infrastructure Compute, storage, networking with soverei...	Operational axis Foundational axis
CIO "Are we using confidential-computing primitives?" 1	Silicon Abstraction Hardware diversity behind portable runtim...	Foundational axis
Board "What's our material AI infrastructure risk for disclosure?" 0	Physical Compute Semiconductors, energy, facilities. The s...	Operational axis
The lens axes · illustrative example		
Threat axis How AI threatens. e.g. Amplifier, Domain, Attacker.	Operational axis How AI integrates. e.g. Defender, Asset, Workforce.	Foundational axis How trust works. e.g. Trustless.
The questions are rotated across the three published lenses (CISO, CIO/CTO, Board) to demonstrate how the same architecture supports different audiences. Each published lens		
Agentics Foundation · OIA Lens System · An integral part of the OIA Reference Model		

One architecture. Three audiences. Each in their own language.

THE SECURITY
ORGANISATION

The CISO Lens

Calibrated to the threat surfaces, frameworks, and planning questions of CISOs in 2026. Three actors to govern: humans, agents, and the foundation models themselves.

FRAMEWORK ANCHORS

NIST CSF · NIST AI RMF · MITRE ATLAS · MITRE D3FEND · OWASP LLM · CCC

THE TECHNOLOGY
ORGANISATION

The CIO/CTO Lens

Calibrated to the conversations about coherence, capital allocation, vendor concentration, and architectural debt that the technology executive is having now.

FRAMEWORK ANCHORS

TOGAF · ITIL · FinOps · ISO/IEC 42001 · platform engineering · SRE

DIRECTORS, RISK
COMMITTEES, CRO

The Board & Risk Lens

Calibrated to accountability and exposure. Treats the cross-layer spans as primary, the layers as substrate. Built to support the conversations directors are actually having.

FRAMEWORK ANCHORS

NIST AI RMF · ISO/IEC 42001 · EU AI Act · SEC disclosure · fiduciary frameworks

WHAT THE LENS WORK REVEALS ABOUT THE ARCHITECTURE

An architecture that holds its shape across audiences is drawn against the structural shape of the problem.

Different audiences foreground different layers. The architecture supports all of them — without distortion.

- Security audience reads Layers 9, 7, 5, 3 as primary.
- Technology audience reads Layers 0–2 as primary, with Layer 8 as the unowned territory.
- Governance audience reads the cross-layer spans as primary, the layers as substrate.

None of these readings distorts the architecture. None requires it to be redrawn to fit.

The Lens Maps are not the proof of the architecture.

The audiences finding themselves on the page is the proof.

WHAT COMES NEXT

The lens roster grows as practitioner engagement supports it.

PUBLISHED IN V0.5

- The CISO Lens
- The CIO/CTO Lens
- The Board and Risk Lens

ANTICIPATED

- The CFO Lens — capital allocation, unit economics
- The General Counsel Lens — liability, regulatory exposure
- The CHRO Lens — agents as workforce, governance

Other lenses are possible — chief data officer, regulated-industry, public-sector, line-of-business.

The Foundation will publish lenses for any audience where the work meets the discipline of the approach. Practitioners with deep audience expertise are invited to contribute proposals.

PART THREE

What This Means for You

Engaging with the OIA across the executive team.

Five artifacts. One coherent release.

OIA Reference Paper

The architectural specification in full. ~40 pages. The authoritative reference.

OIA Lens System

Three audience-specific Lens Maps. CISO, CIO/CTO, Board and Risk.

OIA Visual Reference

Three diagrams that render the architecture and its applications visually.

OIA Body of Work Framing

Reading guide and overview. Where to start by role.

Thought Leadership Deck

This presentation. Calibrated to board and executive audiences.

Different roles. Different entry points. Same architecture.

CISO

The CISO Lens, then Reference Paper §15 (threats) and §14 (spans).

CIO / CTO

The CIO/CTO Lens, then Reference Paper layer chapters most acute to current planning.

Board, Risk Committee, CRO

The Board and Risk Lens, then Reference Paper §14 (spans) and §17 (open questions).

CEO and Operating Executives

This deck. Then the Foreword, Introduction, and Closing of the Reference Paper.

Practitioners and Contributors

Everything. The full Reference Paper, all Lenses, and the Architecture Lock.

The open questions are where the work concentrates next.

Layer 8 — Continuity Fabric.

Continuity primitives that work across agent frameworks. Witness-chain implementations at scale. The most architecturally consequential gap in the present technology stack.

Layer 7 — Agent identity at architectural scope.

Onboarding, scope assignment, behavioural baselining, offboarding. The MCP security architecture as it matures.

Layer 5 — Continuous model trust scoring.

Foundation models as actors whose trustworthiness is dynamic and must be governed dynamically.

Verifiability span.

Whether trust verified rather than asserted becomes an enterprise default — or remains a regulated-workload specialty.

CLOSING

Reference architectures earn their place when the people who must apply them can find their work in the language.

The OIA Reference Model is meant to be foundational. Foundational means open — to the audiences that will use it, to the practitioners who will refine it, and to the technological progress that will keep changing what enterprise intelligence can do.

The Foundation invites you to read the architecture. Read the Lens that fits your role. Push back on what does not fit. Tell us what is missing.

We will keep developing the model. The body of work will keep growing.

WITH THANKS

Acknowledgments

The OIA Reference Model is the product of many minds. The architecture builds on the agentic infrastructure thinking of Reuven Cohen. The pressure-testing and practitioner perspectives came from John McKinley, Greg Lavender, and the CISO colleagues who took the time to read drafts.

The customer and partner perspectives that grounded the architecture in real deployment came from friends at:

Arista Networks · Nutanix · Proofpoint · Reltio · Thoma Bravo · ACIUM

— each contributing technological insights, customer perspectives, and the sage advice that comes from people who have actually had to ship enterprise software at scale. The model is better for all of it.

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Discussion

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